

Bastien Lapierre

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🔍 Google scholar

📄 arXiv

Professional Experience

- 2025 – Now **Junior Research Chair, École Normale Supérieure – PSL**
Postdoctoral fellow of the Philippe Meyer Institute
Independent research program on driven and dissipative quantum phases of matter, with a focus on their theoretical design and relevance to quantum simulation platforms.
- 2023 – 2025 **Postdoctoral Fellow, Princeton University**
Advisor: Shinsei Ryu
Research on entanglement structure in equilibrium and non-equilibrium quantum many-body systems.
- 2019 – 2023 **Doctoral Researcher, University of Zürich**
Advisor: Titus Neupert
PhD research on non-equilibrium quantum critical dynamics and topological phases of disordered systems.

Education

- 2019 – 2023 **Ph.D. Physics, University of Zürich**
Thesis title: *Inhomogeneous and Disordered Quantum Systems: From Dynamics to Topology*
- 2017 – 2019 **M.Sc. Physics, ETH Zürich**
Thesis title: *Heating Dynamics in Floquet Conformal Field Theory*
Honors: Diploma with highest distinction, 5.91/6
- 2014 – 2017 **B.Sc. Physics, University of Geneva**
Honors: Prize for the best B.Sc. in Physics

Skills

Programming	Python, Wolfram Language, Bash
Libraries & Tools	QuSpin, TeNPy, Qiskit, PyTorch, Mathematica. Experience with high-performance computing environments (HPC) and Slurm Workload Manager
Languages	English (fluent), French (mother tongue), German (intermediate), Spanish (basic)

Grants and awards

- 2025 Philippe Meyer Junior Research Chair
- 2023 SNSF Postdoctoral Mobility Fellowship

Grants and awards (continued)

2017 Charles-Eugène Guye Prize (best undergraduate degree in physics)

Mentoring experience

- 2023-2025 Co-supervision of graduate research projects of Zhi-Xing Lin, Princeton University
Outcomes: One peer-reviewed article, one preprint
- 2020-2023 Co-supervision of one M.Sc. thesis (Valerio Pagni) and two B.Sc. theses (Johannes Christmann, Fabian Jaeger)

Teaching experience

- 2020-2023 Teaching assistant for graduate courses at ETH Zürich (Quantum Field Theory I, Advanced Field Theory, General Relativity) and undergraduate courses at the University of Zürich (Analysis I, Linear algebra II, Proseminar of theoretical physics)

Research Publications

Preprints

- Bai, C., Zhou, Z., **Lapierre, B.**, & Ryu, S. (2026). Bootstrapping symmetries in quantum many-body systems from the cross spectral form factor, arxiv:2604.01296.
- Rhyno, B., **Lapierre, B.**, Vishveshwara, S., Najafi, K., & Chitra, R. (2026). Observing quantum many-body dynamics in emergent curved spacetime using programmable quantum processors, arxiv:2602.15524.
- Lapierre, B.**, Mo, L.-H., & Ryu, S. (2025). Entanglement transitions in structured and random nonunitary gaussian circuits, arXiv:2507.03768.
- Lapierre, B.**, Moosavi, P., & Oblak, B. (2025). Nonequilibrium probes of quantum geometry in gapless systems, arxiv:2511.09639.
- Lin, Z.-X., Prem, A., Ryu, S., & **Lapierre, B.** (2025). Local-to-global entanglement dynamics by periodically driving impurities. arXiv:2510.20908.

Journal Articles

- Bai, C., Tian, M. T., **Lapierre, B.**, & Ryu, S. (2026). Spatially structured entanglement from nonequilibrium thermal pure states. *Phys. Rev. B*, *113*, 064311.
- Lapierre, B.**, Trifunovic, L., Neupert, T., & Brouwer, P. W. (2026). Topology of ultralocalized insulators and superconductors. *Phys. Rev. Lett.*, *136*, 066601.
- Lapierre, B.**, Numasawa, T., Neupert, T., & Ryu, S. (2025). Floquet engineered inhomogeneous quantum chaos in critical systems. *Phys. Rev. B*, *112*, 104317.
- Lapierre, B.**, Pelliconi, P., Ryu, S., & Sonner, J. (2025). Driven nonunitary dynamics of quantum critical systems. *Phys. Rev. B*, *112*, 104322.
- Moosavi, P., Oblak, B., **Lapierre, B.**, Estienne, B., & Stéphan, J.-M. (2025). Quantum hall edges beyond the plasma analogy. *Phys. Rev. Res.*, *7*, 023134.

- Lin, Z.-X., **Lapierre, B.**, Moosavi, P., & Ryu, S. (2025). Chiral instabilities in driven-dissipative quantum liquids. *Phys. Rev. B*, *111*, 165131, Editors' Suggestion.
- Oblak, B., **Lapierre, B.**, Moosavi, P., Stéphan, J.-M., & Estienne, B. (2024). Anisotropic quantum Hall droplets. *Phys. Rev. X*, *14*, 011030.
- Datta, S., **Lapierre, B.**, Moosavi, P., & Tiwari, A. (2023). Marginal quenches and drives in Tomonaga-Luttinger liquids. *SciPost Phys.*, *14*, 108.
- Molignini, P., **Lapierre, B.**, Chitra, R., & Chen, W. (2023). Probing Chern number by opacity and topological phase transition by a nonlocal Chern marker. *SciPost Phys. Core*, *6*, 059.
- Lapierre, B.**, Neupert, T., & Trifunovic, L. (2022). Topologically localized insulators. *Phys. Rev. Lett.*, *129*, 256401.
- Choo, K., **Lapierre, B.**, Kuhlenskamp, C., Tiwari, A., Neupert, T., & Chitra, R. (2022). Thermal and dissipative effects on the heating transition in a driven critical system. *SciPost Physics*, *13*(5).
- Lapierre, B.**, Neupert, T., & Trifunovic, L. (2021). *N*-band Hopf insulator. *Phys. Rev. Research*, *3*, 033045.
- Lapierre, B.**, & Moosavi, P. (2021). Geometric approach to inhomogeneous Floquet systems. *Phys. Rev. B*, *103*, 224303.
- Lapierre, B.**, Choo, K., Tiwari, A., Tauber, C., Neupert, T., & Chitra, R. (2020). Fine structure of heating in a quasiperiodically driven critical quantum system. *Phys. Rev. Research*, *2*, 033461.
- Lapierre, B.**, Choo, K., Tauber, C., Tiwari, A., Neupert, T., & Chitra, R. (2020). Emergent black hole dynamics in critical Floquet systems. *Phys. Rev. Research*, *2*, 023085.

Conference talks

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| 12/2025 | Workshop on Beyond Linearity: Frontiers in Non-Hermitian Many-Body Physics, Erice, Italy, “ <i>Driven non-unitary dynamics of quantum critical systems</i> ” |
| 06/2024 | Workshop on Non-equilibrium Many-body Physics Beyond the Floquet Paradigm, Max Planck Institute for the Physics of Complex Systems, Germany, “ <i>Fractal entanglement transitions in a quasiperiodic non-unitary circuit</i> ” |
| 03/2024 | APS March Meeting 2024, Minneapolis, USA, “ <i>Anisotropic Quantum Hall Droplets</i> ”. |
| 07/2023 | Workshop on Mathematical Aspects of Condensed Matter Physics, ETH Zürich, Switzerland, “ <i>Topologically localized phases</i> ” |
| 06/2022 | Workshop on Out-of-equilibrium and collective dynamics of quantum many-body systems, ETH Zürich, Switzerland, “ <i>Marginal quenches and drives in Tomonaga-Luttinger liquids</i> ” |
| 06/2021 | Workshop on Low dimensional quantum many-body systems, Heidelberg, Germany, “ <i>Geometry and black holes in periodically driven critical quantum systems</i> ” |
| 03/2021 | APS March Meeting 2021, online, “ <i>Geometry and black holes in periodically driven critical quantum systems</i> ” |

Invited Seminars

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| 02/2026 | Statistical Physics Forum, Ecole Normale Supérieure – PSL, France |
| 11/2025 | Philippe Meyer Institute Colloquium, Ecole Normale Supérieure – PSL, France |
| 09/2025 | Condensed Matter & AMO Seminar, Georgia Institute of Technology, USA |

Invited Seminars (continued)

07/2025	Theoretical Physics Seminar, Kavli Institute for Theoretical Sciences, University of Chinese Academy of Sciences, China
01/2025	Condensed Matter Seminar, LPTM, CY Cergy Paris Université, France
08/2024	Group Seminar, Collège de France, France
07/2024	Condensed Matter Seminar, LPMMC, Université Grenoble Alpes, France
06/2024	Group Seminar, European Center for Quantum Sciences, Université de Strasbourg, France
05/2024	Leeds-Loughborough-Nottingham Non-Equilibrium Seminar Series, United Kingdom
04/2024	Mathematical Physics Seminar, Université de Montréal, Canada
02/2024	Theoretical Physics Seminar, École Normale Supérieure de Lyon, France
11/2023	Quantum Initiative Seminar, Princeton University, USA
01/2023	Condensed Matter Seminar, Freie Universität Berlin, Germany
10/2021	Speakers' Corner, online
04/2021	Condensed Matter Seminar, IFW Dresden, Germany
10/2019	Condensed Matter Seminar, University of Zürich, Switzerland
06/2019	Group Seminar, ETH Zürich, Switzerland

Scientific outreach

06/2024	Jury at Princeton Research Day 2024: graded the research projects of undergraduate and graduate students, and gave them feedback on their scientific communication skills.
03/2024	Outreach talk at Princeton Postdoctoral Council Seminar Series, " <i>Emergent Black Hole Dynamics in Quantum Matter</i> " Audience: Postdocs from Princeton University across all departments Awarded best talk of the semester
11/2022	Outreach talk on the 2022 Physics Nobel Prize, as part of the nanoTalks series from Reatch, Zürich